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Faculty of Science, Agriculture and Engineering

ASSIGNMENT COVER SHEET

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Abstract

This report presents a comprehensive power system study that includes power-flow preparation, load flow analysis, and fault analysis using the PowerWorld Simulator. In the preparation phase, all system parameters, including bus data, transmission lines, and transformers, were converted into per-unit values based on 100 MVA and 13.8 kV bases, with particular attention to Line 2 (5.3 km). The load flow analysis was then carried out using the Gauss-Seidel method to establish bus voltages, line flows, and system losses under normal conditions. Different scenarios were tested, such as raising the voltage at Bus 4 by adding a shunt capacitor, and doubling the load at Buses 3 to 5, which required transformer tap adjustments and additional reactive support to restore voltage within acceptable limits. Finally, a fault analysis was performed on the seven-bus system, combining digital simulation and analytical hand calculations. Sub-transient fault currents and voltages were obtained for three-phase faults at each bus, and the results for Bus 4 showed strong agreement between simulation and analytical methods. Overall, the study highlights the importance of per-unit normalization, reactive power control, and fault analysis in ensuring reliable, efficient, and secure operation of power systems.

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Introduction

The reliable operation of a power system depends on its ability to maintain stability, manage load flow, and withstand abnormal conditions such as faults [1]. This report presents the analysis of a radial distribution feeder system, focusing on power flow studies and fault analysis using the PowerWorld Simulator [2]. The single-line diagram includes a generator, transformer, transmission lines, and several load buses with specified real and reactive power demands, where Bus 1 is defined as the swing bus to provide a voltage reference [3].

The first stage of the study involves converting all network parameters, including impedances, voltages, and loads into per-unit values based on standard base quantities ($S_{base} = 10 \text{ MVA}$, $V_{base} = 13.8 \text{ kV}$) [4]. These values are organized into input tables for buses, lines, and transformers, forming the foundation for steady-state load flow analysis [5]. The power flow simulations explore system behaviour under different operating conditions, such as targeted voltage improvement at a specific bus and the impact of a 100% load increase, while assessing changes in losses and voltage profiles [6].

In addition, fault analysis is performed to evaluate the system response under abnormal short-circuit conditions, particularly three-phase faults [7]. This includes both digital simulations and analytical hand calculations, such as determining the Thevenin equivalent impedance and fault currents at Bus 4 [8]. The results are compared to validate the accuracy of both approaches and to emphasize the importance of predictive fault studies in designing protective equipment and ensuring secure grid operation [9].

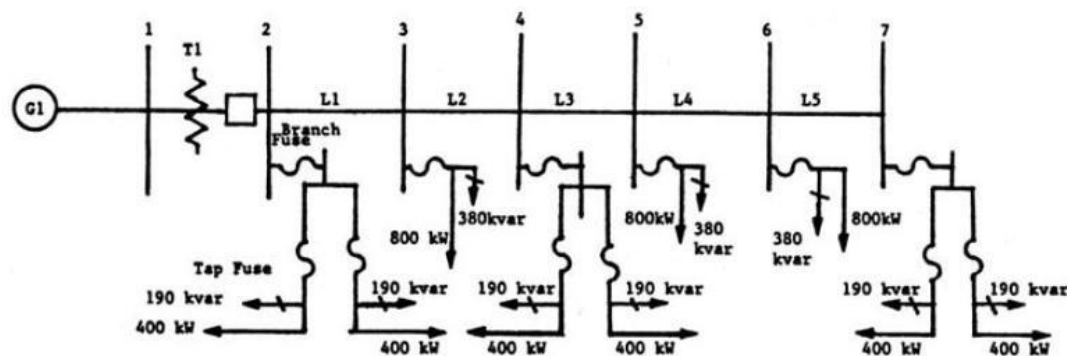


Figure 1: Single-line diagram.

Task 1: tables and calculations.

This task involves converting all positive-sequence impedance, load, and voltage data to per unit using the given system base quantities: $S_{base} = 10 \text{ MVA}$ (three-phase) and $V_{base} = 13.8 \text{ kV}$ (line-to-line) in the zone of the lines.

Equations:

$$X_{p.u.} = X_{p.u.o} \left(\frac{V_{b.o}}{V_{b.n}} \right)^2 \frac{S_{b.n}}{S_{b.o}} \quad (1)$$

$$Z_{base} = \frac{(V_{base})^2}{S_{base}} \quad (2)$$

$$Y_{base} = \frac{1}{Z_{base}} \quad (3)$$

$$Z_{p.u.} = \frac{Z_{nominal}}{Z_{base}} \quad (4)$$

- System (feeder) base: $S_{base} = 10 \text{ MVA}$, $V_{base, LL} = 13.8 \text{ kV}$.
- Base impedance at 13.8 kV:

$$Z_{base, 13.8} = \frac{V_{base}^2}{S_{base}} = \frac{(13.8 \times 10^3)^2}{10 \times 10^6} = 19.044 \, \Omega$$

- For the 345 kV side (used for G1):

$$Z_{base, 345} = \frac{(345 \times 10^3)^2}{10 \times 10^6} = 11\,902.5 \, \Omega$$

Base admittance: $Y_{base} = 1/Z_{base} = 0.05251 \text{ S}$

X_{new} for G1 (generator)

Given (from figure):

- $X' = 0.15 \text{ pu}$ on a 50 MVA machine base. $S_{new} = 10 \text{ MVA}$

$$X_{new} = X_{old} \times \frac{S_{new}}{S_{old}}$$

$$X_{new} = 0.15 \times \frac{10}{50} = 0.15 \times 0.2 = \mathbf{0.03 \text{ pu}}$$

X_{new} for T1 (transformer)

Given: transformer reactance $X_T = 0.10$ pu on a 5 MVA transformer base (345/13.8 kV)

$$X_{T,new} = 0.10 \times \frac{10}{5} = 0.10 \times 2 = \mathbf{0.20 \text{ pu}}$$

Lines L1 → L5 impedance

Given line data (per km):

- $R = 0.19 \text{ } \Omega/\text{km}$
- $X = 0.38 \text{ } \Omega/\text{km}$

- $R_{pu_per_km} = \frac{0.19}{19.044} \approx 0.00998 \text{ pu/km}$
- $X_{pu_per_km} = \frac{0.38}{19.044} \approx 0.01995379 \text{ pu/km}$

- Positive-seq series impedance per km:

$$z_1 = 0.19 + j0.38 \text{ } \Omega/\text{km}$$

- Shunt admittance per km:

$$y_1 = j 4.0 \times 10^{-6} \text{ S/km}$$

$$y_{pu/km} = y_1 Z_{base} = j(4.0 \times 10^{-6})(19.044) = j 7.6176 \times 10^{-5} \text{ pu/km}$$

$$B_{pu/km} = 7.6176 \times 10^{-5} \text{ pu/km}$$

$$B_{pu} = \text{Im}(y_{pu/km}) \cdot \text{length}$$

Line Lengths: L1=L3=L4=L5= 2.0km and L2= 5.3km

1. Impedance of Line L1 (Z_{L1})

$$Z_{L1} = z_{pu/km} \times L1$$

$$Z_{L1} = (0.00998 + j0.01995379) \times 2.0$$

- Resistance (R_{L1}): $0.00998 \times 2.0 = \mathbf{0.01996}$ pu
- Reactance (X_{L1}): $0.01995379 \times 2.0 = \mathbf{0.03990758}$ pu

$$\mathbf{Z_{L1} \approx 0.01996 + j0.03990758 \text{ pu}}$$

2. Impedance of Line L2 (Z_{L2})

$$Z_{L2} = z_{pu/km} \times L2$$

$$Z_{L2} = (0.00998 + j0.01995379) \times 5.3$$

- Resistance (R_{L2}): $0.00998 \times 5.3 = \mathbf{0.052894}$ pu
- Reactance (X_{L2}): $0.01995379 \times 5.3 = \mathbf{0.105755087}$ pu

$$\mathbf{Z_{L2} \approx 0.052894 + j0.105755087 \text{ pu}}$$

Loads (per-unit)

GIVEN:

- 400 kW load $\rightarrow P = 400 \text{ kW}$
- 800 kW load $\rightarrow P = 800 \text{ kW}$

On the 10 MVA base:

$$P_{pu}(400 \text{ kW}) = \frac{0.400 \text{ MW}}{10 \text{ MVA}} = \mathbf{0.04 \text{ pu}}$$

$$P_{pu}(800 \text{ kW}) = \frac{0.800 \text{ MW}}{10 \text{ MVA}} = \mathbf{0.08 \text{ pu}}$$

Reactive capacitors (capacitors supply reactive $\rightarrow$ negative Q in load convention):

$$Q_{pu}(190 \text{ kvar}) = -\frac{0.190 \text{ Mvar}}{10 \text{ MVA}} = \mathbf{-0.019 \text{ pu}}$$

$$Q_{pu}(380 \text{ kvar}) = -\frac{0.380 \text{ Mvar}}{10 \text{ MVA}} = \mathbf{-0.038 \text{ pu}}$$

Voltages

Bus 1: $V = 1.00 \angle 0^\circ \text{ pu}$ (swing)

Table 1: Impedances and per-unit values of transmission lines

Line	Length (km)	$Z [\Omega]$	Z_{pu}
L1	2	$0.38 + 0.76j$	$0.02 + 0.04j$
L2	5.3	$0.665 + 2.014j$	$0.035 + 0.106j$
L3	2	$0.38 + 0.76j$	$0.02 + 0.04j$
L4	2	$0.38 + 0.76j$	$0.02 + 0.04j$
L5	2	$0.38 + 0.76j$	$0.02 + 0.04j$

Loads

$$S_{p.u} = \frac{(0.0800 + 0.0380j)MVA}{10MVA}$$

$$= 0.08 + 0.038j \text{ p.u}$$

Table 2: Bus Input Data.

Bus	Bus Type	V	δ°	P_G	Q_G	P_L	Q_L	Q_{max}	Q_{min}
p.u	Deg.	MW		MVar	MW	MW	MVar	MVar	MVar
1	Swing	1.0	0	-	-	0	0	-	-
2	Load	-	-	0	0	800	380	-	-
3	Load	-	-	0	0	800	380	-	-
4	Load	-	-	0	0	800	380	-	-
5	Load	-	-	0	0	800	380	-	-
6	Load	-	-	0	0	800	380	-	-
7	Load	-	-	0	0	800	380	-	-

Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW	Load Mvar
1	1	1	345.00	1.00000	345.000	0.00	3.62	1.72		
2	2	1	13.80	0.99659	13.753	-0.42			0.40	0.19
3	3	1	13.80	0.99535	13.736	-0.47			0.80	0.38
4	4	1	13.80	0.99331	13.708	-0.60			0.40	0.19
5	5	1	13.80	0.99253	13.697	-0.64			0.80	0.38
6	6	1	13.80	0.99207	13.691	-0.66			0.80	0.38
7	7	1	13.80	0.99191	13.688	-0.66			0.40	0.19
12	12	1	138.00	0.00000	0.000	0.00				

Table 3: Transmission Lines Input Data

Bus-to-Bus	R' [p.u]	X' [p.u]	G' [p.u]	B' [p.u]	MVA_{max} [p.u]
2-3	0.02	0.04	-	0.000152	5
3-4	0.035	0.106	-	0.00042	0.2
4-5	0.02	0.04	-	0.000152	0.5
5-6	0.02	0.04	-	0.000152	0.5
6-7	0.02	0.04	-	0.000152	0.5

From Number	From Name	To Number	To Name	Circuit	Status	Branch Device Type	Xtrmr	R	X	B	Lim MVA A	Lim MVA B	Lim MVA C
1	1 1	2 2	2 2	1	Closed	Transformer	YES	0.00000	0.20000	0.00000	5.0	0.0	0.0
2	6 6	7 7	7 7	1	Closed	Line	NO	0.02000	0.04000	0.00001	5.0	0.0	0.0
3	4 4	5 5	5 5	1	Closed	Line	NO	0.02000	0.04000	0.00001	5.0	0.0	0.0
4	5 5	6 6	6 6	1	Closed	Line	NO	0.02000	0.04000	0.00001	5.0	0.0	0.0
5	2 2	3 3	3 3	1	Closed	Line	NO	0.02000	0.04000	0.00001	5.0	0.0	0.0
6	3 3	4 4	4 4	1	Closed	Line	NO	0.02500	0.10600	0.00002	5.0	0.0	0.0

Table 4: Transformer Input Data

Bus-to-Bus	R [p.u]	X [p.u]	G_c [p.u]	B_m [p.u]	MVA_{max} [p.u]	Max TAP setting [p.u]
1-2	-	0.2			0.2	-

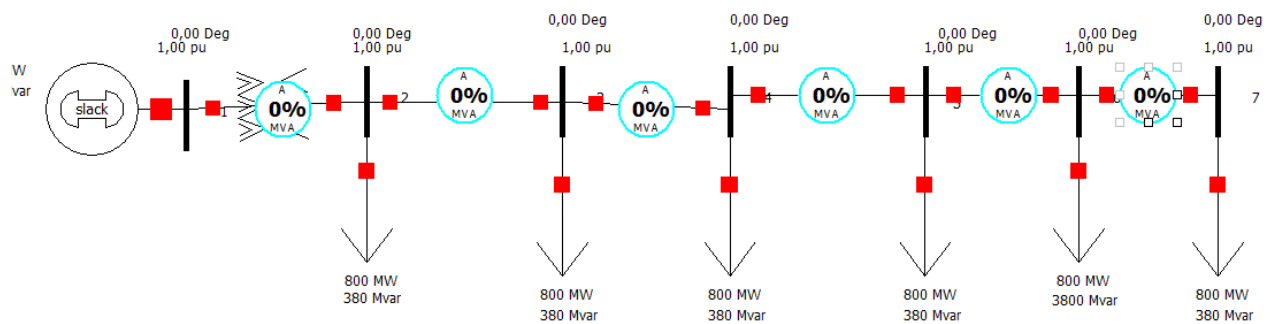


Figure 2 Circuit before running power flow.

Task 2

Case 1

Circuit Diagrams and Tables

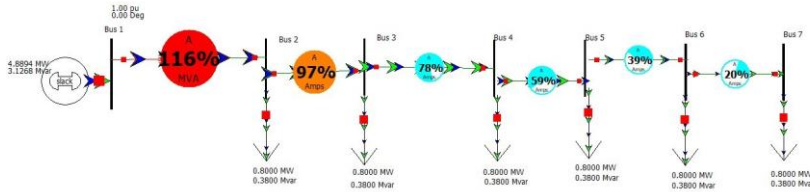


Figure 3 One Line Diagram After Gauss-Seidel Solver

Bus	Nominal Voltage	V [p.u]	Voltage [kV]	δ°	P_L [MW]	Q_L [MVar]	P_G [MW]	Q_G [MVar]
1	345	1.0	345	0	-	-	4.89	3.13
2	13.8	0.94255	13.007	-5.96	0.8	0.38	-	-
3	13.8	0.92520	12.768	-6.76	0.8	0.38	-	-
4	13.8	0.91827	12.627	-7.08	0.8	0.38	-	-
5	13.8	0.90786	12.528	-7.59	0.8	0.38	-	-
6	13.8	0.90092	12.433	-7.93	0.8	0.38	-	-
7	13.8	0.89745	12.385	-8.10	0.8	0.38	-	-

Table 5: Bus Output Data Task 2 Case 1

Buses										
Filter Advanced Bus Find... Remove Quick Filter										
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW
1	1	1	1	345.00	1.00000	345.000	0.00	3.62	1.72	
2	2	2	1	13.80	0.99658	13.753	-0.42			
3	3	3	1	13.80	0.99533	13.736	-0.47			
4	4	4	1	13.80	0.99329	13.707	-0.60			
5	5	5	1	13.80	0.99251	13.697	-0.63			
6	6	6	1	13.80	0.99204	13.690	-0.66			
7	7	7	1	13.80	0.99189	13.688	-0.66			
8	12	12	1	138.00	0.00000	0.000	0.00			

Table 6: Transformer and Line Output Data

Branch											
Filter Advanced Branch Find... Remove Quick Filter											
	From Number	From Name	To Number	To Name	Circuit	Status	Branch Device Type	Xfrmr	MW From	Mvar From	MVA
1	1	1	2	2	1	Closed	Transformer	YES	3.6	1.7	
2	2	2	3	3	1	Closed	Line	NO	3.2	1.5	
3	3	3	4	4	1	Closed	Line	NO	2.4	1.1	
4	4	4	5	5	1	Closed	Line	NO	2.0	0.9	
5	5	5	6	6	1	Closed	Line	NO	1.2	0.6	
6	6	6	7	7	1	Closed	Line	NO	0.4	0.2	

Case 2

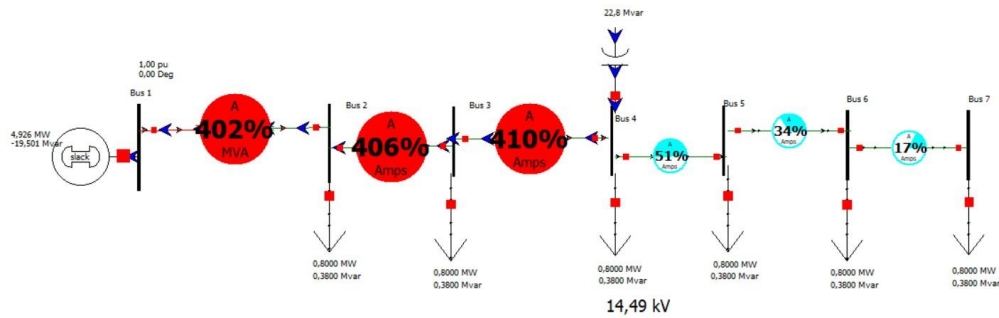


Figure 4: : Method 1 To Increase Voltage At Bus 4 by 5%

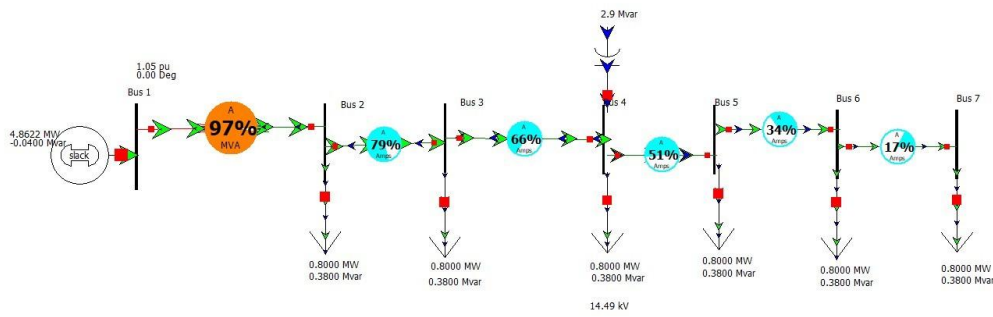


Figure 5: Method 2 To Increase Voltage At Bus 4 by 5%

Two methods were tried to increase the voltage at bus 4 by 5%. The first method was to add a shunt capacitor at Bus 4, but this overloads the Transformer and the transmission lines from bus 2 to bus 4. In the second method, By increasing the Voltage of the Generator at the Slack bus from 1.0 p.u to 1.05 p.u, the voltage at Bus 4 will also increase by 5%, but since the transformer overloads, a shunt capacitor is added at bus 4 to compensate for the reactive power. The output data tables can be seen below

Table 7: Bus Output Data

Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW	Load Mvar	Dist MW	Dist Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num
1	1	1	345.00	1.00000	345.000	0.00	3.73	-14.58						0.00	0.00	1
2	2	1	13.80	1.02918	14.203	-0.42			0.40	0.19	0.00	0.00		0.00	0.00	1
3	3	1	13.80	1.03446	14.276	-0.65			0.80	0.38	0.00	0.00		0.00	0.00	1
4	4	1	13.80	1.04973	14.486	-1.08			0.40	0.19	0.00	0.00	17.08	0.00	0.00	1
5	5	1	13.80	1.04899	14.476	-1.11			0.80	0.38	0.00	0.00		0.00	0.00	1
6	6	1	13.80	1.04854	14.470	-1.13			0.80	0.38	0.00	0.00		0.00	0.00	1
7	7	1	13.80	1.04839	14.468	-1.14			0.40	0.19	0.00	0.00		0.00	0.00	1
12	12	1	138.00	0.00000	0.000	0.00								0.00	0.00	1

Table 8: Transformer Output Data

Branches State X Buses X Loads

Records

Geo

Set

Columns

</

Firstly, target voltage was calculated.

- A 5% increase means new target is $V4_new = 13.8\text{kV} * 1.05 = 14.49\text{kV}$.

A shunt capacitor bank was then added at Bus 4. Capacitors inject reactive power into the system, which reduces the reactive current flow in the lines and consequently reduces the voltage drop, thereby raising the voltage at the bus.

The capacitor size was adjusted until the voltage at bus 4 is as close to the target (14.49kV) as possible.

Case 3

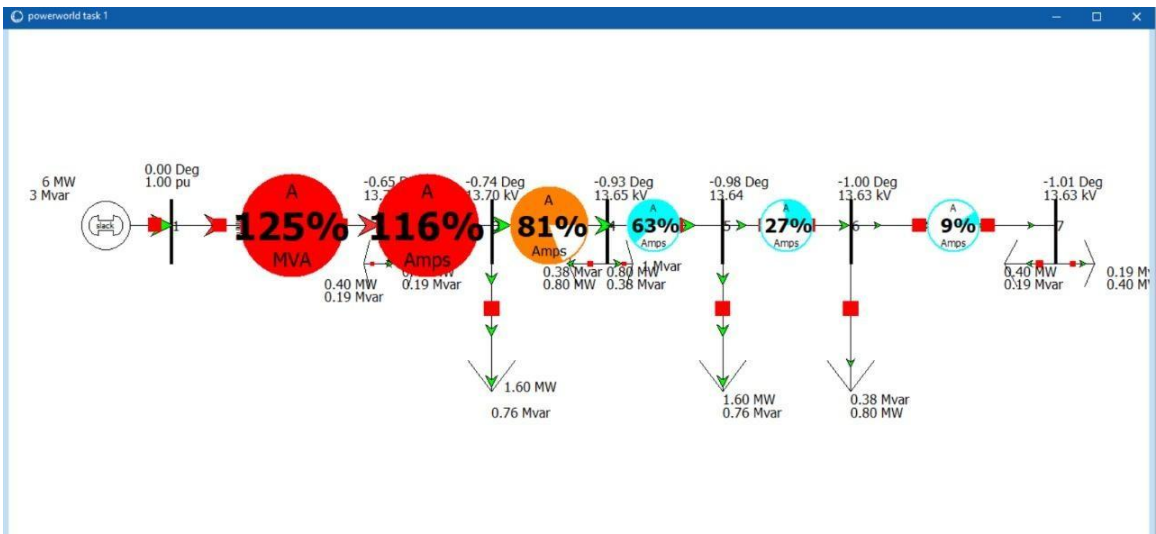


Figure 5: Loads Increased by 100% At Bus 3-5

Table 9: Bus output data.

X Buses X Branches State X Loads																	
Records • Geo • Set • Columns • Options																	
Filter Advanced • Bus • Find... Remove Quick Filter •																	
	Number	Name	Area Name	Nom KV	PU Volt	Volt (KV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW	Load Mvar	Dist MW	Dist Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num
1	1	1	1	345.00	1.00000	345.000	0.00	5.62	2.76						0.00	0.00	1
2	2	2	1	13.80	0.99455	13.725	-0.65			0.40	0.19	0.00	0.00		0.00	0.00	1
3	3	3	1	13.80	0.99250	13.696	-0.74			1.60	0.76	0.00	0.00		0.00	0.00	1
4	4	4	1	13.80	0.98939	13.654	-0.93			0.80	0.38	0.00	0.00		0.00	0.00	1
5	5	5	1	13.80	0.98829	13.638	-0.98			1.60	0.76	0.00	0.00		0.00	0.00	1
6	6	6	1	13.80	0.98782	13.632	-1.00			0.80	0.38	0.00	0.00		0.00	0.00	1
7	7	7	1	13.80	0.98766	13.630	-1.01			0.40	0.19	0.00	0.00		0.00	0.00	1

Table 10: Line and transformer output data.

	From Number	From Name	To Number	To Name	Circuit	Status	Branch Device Type	Xfrmr	MW From	Mvar From	MVA From	Lim MVA	% of MVA Limit (Max)	MW Loss	Mvar Loss
1	1		2	2	1	Closed	Transformer	YES	5.6	2.8	6.3	5.0	125.2	0.00	0.08
2	2		3	3	1	Closed	Line	NO	5.2	2.5	5.8	5.0	115.7	0.01	0.01
3	3		4	4	1	Closed	Line	NO	3.6	1.7	4.0	5.0	80.0	0.01	0.02
4	4		5	5	1	Closed	Line	NO	2.8	1.3	3.1	5.0	62.0	0.00	0.00
5	5		6	6	1	Closed	Line	NO	1.2	0.6	1.3	5.0	26.6	0.00	-0.00
6	6		7	7	1	Closed	Line	NO	0.4	0.2	0.4	5.0	8.9	0.00	-0.00

The Loads were doubled

-In the Case 1 model (without the capacitor from Case 2), the active (MW) and reactive (MVAR) were doubled at Buses 3, 4, and 5.

Then run the analysis and observe the results.

Losses were compared and the system losses (sum of losses from all lines and the transformers on the Line Data table) were significantly higher than in Case 1. This is due to the squared relationship between current and loss ($P_{loss} = I^2R$); doubling the load current approximately quadruples the losses.

The bus voltages were found to be within $\pm 5\%$ of the nominal value.

Task3

Case1

To perform fault analysis, the input data from Tables 7, 8, and 9 were used.

Bus	$X_2[p.u]$	$X_1[p.u]$	$X_0[p.u]$
1	1	1	1

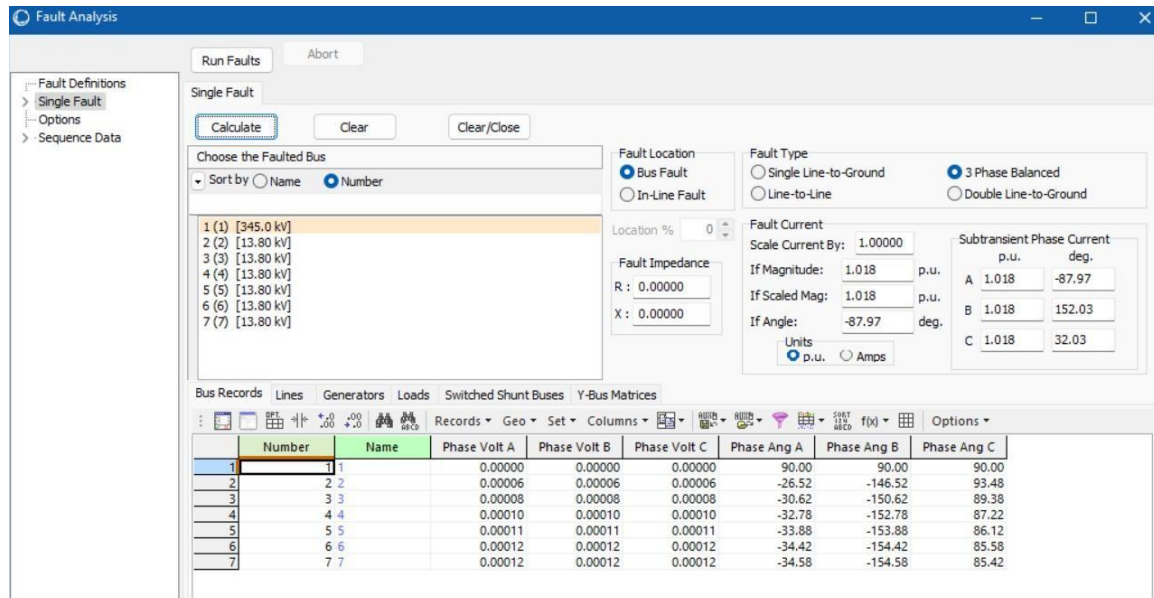


Figure 6: Fault at bus1

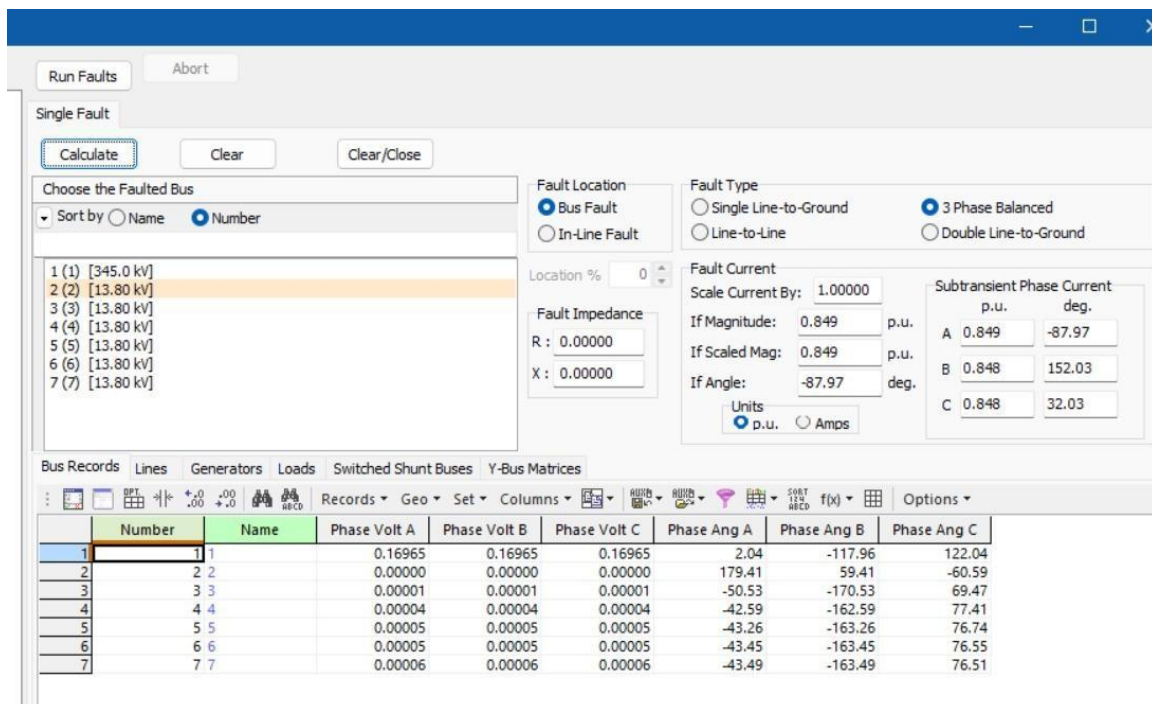


Figure 7: Fault at bus 2

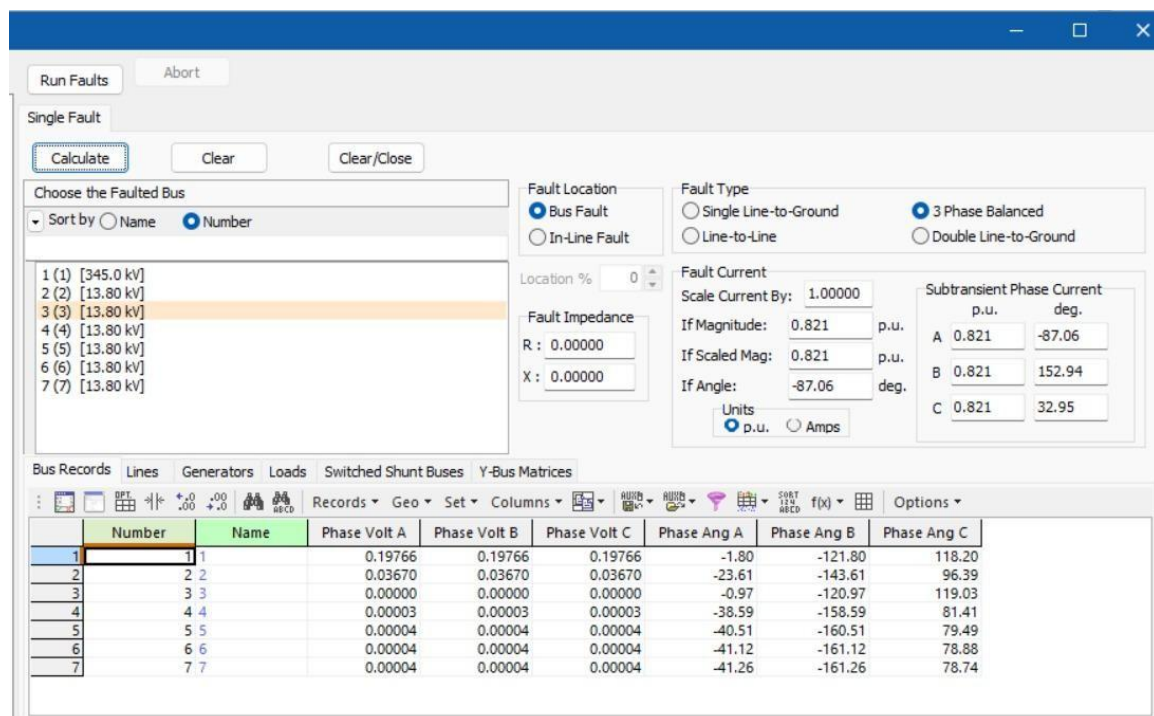


Figure 8: Fault at bus 3

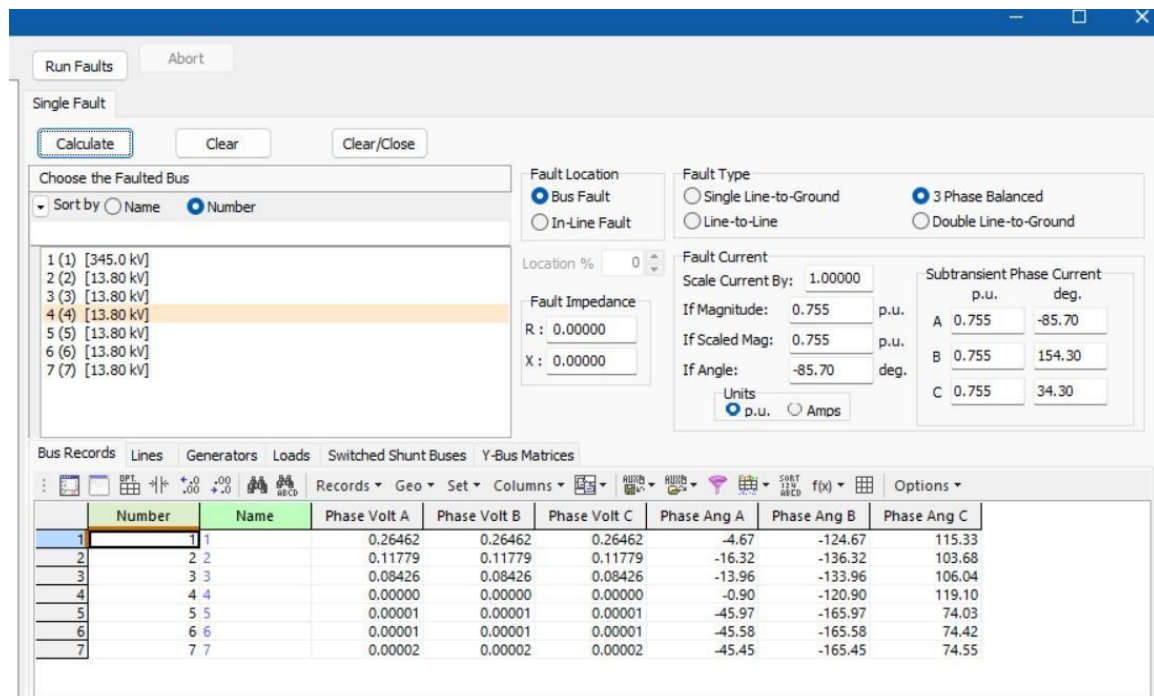


Figure 9: Fault at bus 4

Run Faults Abort

Single Fault

Calculate Clear Clear/Close

Choose the Faulted Bus

Sort by ☐ Name ☒ Number

1 (1) [345.0 kV]
 2 (2) [13.80 kV]
 3 (3) [13.80 kV]
 4 (4) [13.80 kV]
 5 (5) [13.80 kV]
 6 (6) [13.80 kV]
 7 (7) [13.80 kV]

Fault Location
☒ Bus Fault
☐ In-Line Fault

Location % 0

Fault Impedance
 R : 0.00000
 X : 0.00000

Fault Type
☐ Single Line-to-Ground
☐ Line-to-Line
☒ 3 Phase Balanced
☐ Double Line-to-Ground

Fault Current
 Scale Current By: 1.00000
 If Magnitude: 0.732 p.u.
 If Scaled Mag: 0.732 p.u.
 If Angle: -84.97 deg.

Subtransient Phase Current
 p.u. deg.
 A 0.732 -84.97
 B 0.732 155.03
 C 0.732 35.03

Units
☒ p.u. ☐ Amps

Bus Records Lines Generators Loads Switched Shunt Buses Y-Bus Matrices

	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	1	0.28819	0.28819	0.28819	-5.89	-125.89	114.11
2	2	2	0.14690	0.14690	0.14690	-16.90	-136.90	103.10
3	3	3	0.11426	0.11426	0.11426	-15.60	-135.60	104.40
4	4	4	0.03275	0.03275	0.03275	-21.53	-141.53	98.47
5	5	5	0.00000	0.00000	0.00000	-176.52	63.48	-56.52
6	6	6	0.00001	0.00001	0.00001	-44.92	-164.92	75.08
7	7	7	0.00001	0.00001	0.00001	-44.79	-164.79	75.21

Auto Insert... Load Data... Save Data... Close Help

Figure 10: Fault at bus 5

Run Faults Abort

Single Fault

Calculate Clear Clear/Close

Choose the Faulted Bus

Sort by ☐ Name ☒ Number

1 (1) [345.0 kV]
 2 (2) [13.80 kV]
 3 (3) [13.80 kV]
 4 (4) [13.80 kV]
 5 (5) [13.80 kV]
 6 (6) [13.80 kV]
 7 (7) [13.80 kV]

Fault Location
☒ Bus Fault
☐ In-Line Fault

Location % 0

Fault Impedance
 R : 0.00000
 X : 0.00000

Fault Type
☐ Single Line-to-Ground
☐ Line-to-Line
☒ 3 Phase Balanced
☐ Double Line-to-Ground

Fault Current
 Scale Current By: 1.00000
 If Magnitude: 0.711 p.u.
 If Scaled Mag: 0.711 p.u.
 If Angle: -84.29 deg.

Subtransient Phase Current
 p.u. deg.
 A 0.711 -84.29
 B 0.711 155.71
 C 0.711 35.71

Units
☒ p.u. ☐ Amps

Bus Records Lines Generators Loads Switched Shunt Buses Y-Bus Matrices

	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	1	0.31059	0.31059	0.31059	-6.76	-126.76	113.24
2	2	2	0.17434	0.17434	0.17434	-17.05	-137.05	102.95
3	3	3	0.14258	0.14258	0.14258	-16.23	-136.23	103.77
4	4	4	0.06358	0.06358	0.06358	-20.84	-140.84	99.16
5	5	5	0.03178	0.03178	0.03178	-20.85	-140.85	99.15
6	6	6	0.00000	0.00000	0.00000	2.43	-117.57	122.43
7	7	7	0.00000	0.00000	0.00000	-44.12	-164.12	75.88

Figure 11: Fault at bus 6

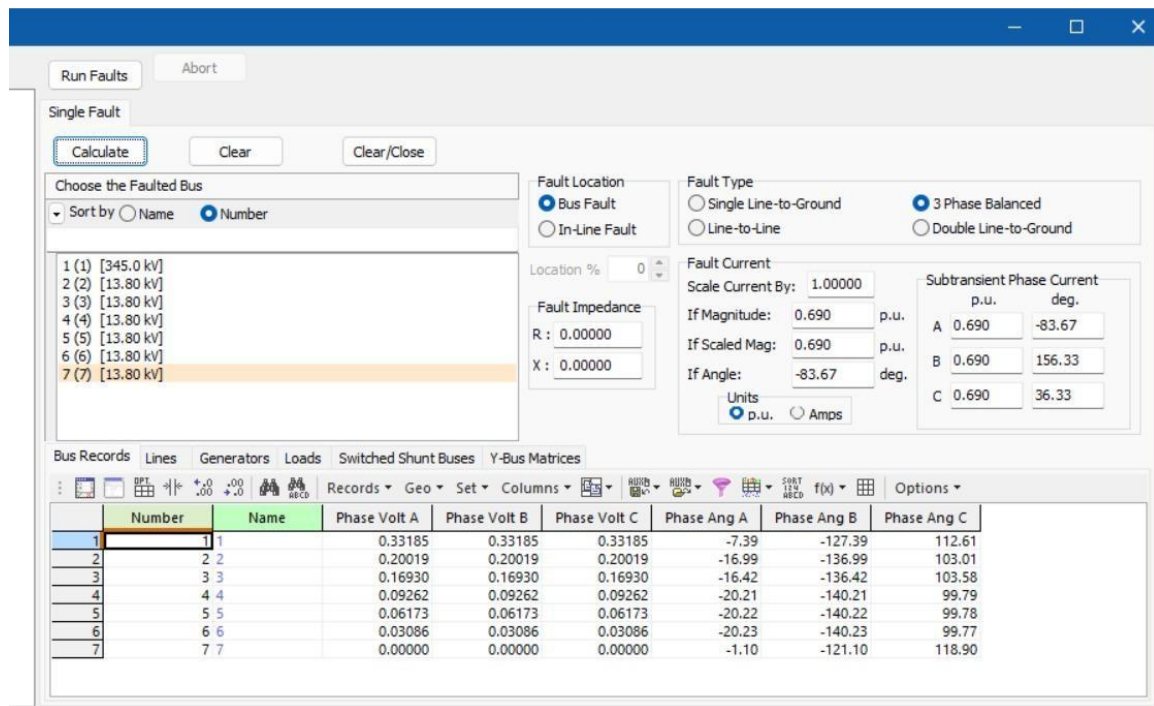


Figure 12: Fault at bus 7

Case2

$$Z_{th} = Z_G + Z_T + Z_{L1} + Z_{L2} = 0.055 + 0.376j \text{ p.u.} = 0.380 \angle 81.678^\circ$$

$$V_{th} = 1 \angle 0^\circ$$

$$I_f = 2.632 \angle -81.678^\circ$$

The hand calculation assumed no current from L_3 to L_5 , making $V_5 = 0$

$$V_3 = 0$$

Discussions and Conclusions

This study combined power-flow preparation, load flow analysis, and fault analysis to build a complete understanding of the electrical network [1]. The preparation phase, which involved converting all parameters into per-unit values, proved essential for simplifying calculations, ensuring consistency across buses, lines, and transformers, and providing reliable input data for simulations [2].

The power flow analysis demonstrated the importance of voltage control and loss management [3]. The base case served as a benchmark, while further scenarios highlighted effective solutions such as shunt capacitor banks for localized voltage support and transformer tap adjustments for system-wide regulation under stressed conditions [4]. The significant increase in losses during the 100% load rise emphasized the non-linear impact of current on efficiency, showing how proactive system management and reactive power compensation are vital for maintaining stability and keeping voltages within the statutory $\pm 5\%$ range [5].

The fault analysis confirmed that both hand calculations and PowerWorld simulations are grounded in the same principles [6]. While minor discrepancies occurred due to rounding in manual work versus the precision of digital methods, both approaches complemented each other [7]. Hand calculations strengthened theoretical understanding, while simulations provided accurate and practical insights into real system performance [8].

Overall, the project shows that integrating per-unit normalization, load flow studies, and fault analysis provides engineers with the tools needed to safeguard grid reliability, ensure operational efficiency, and design effective control and protection strategies for modern power systems [9].

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